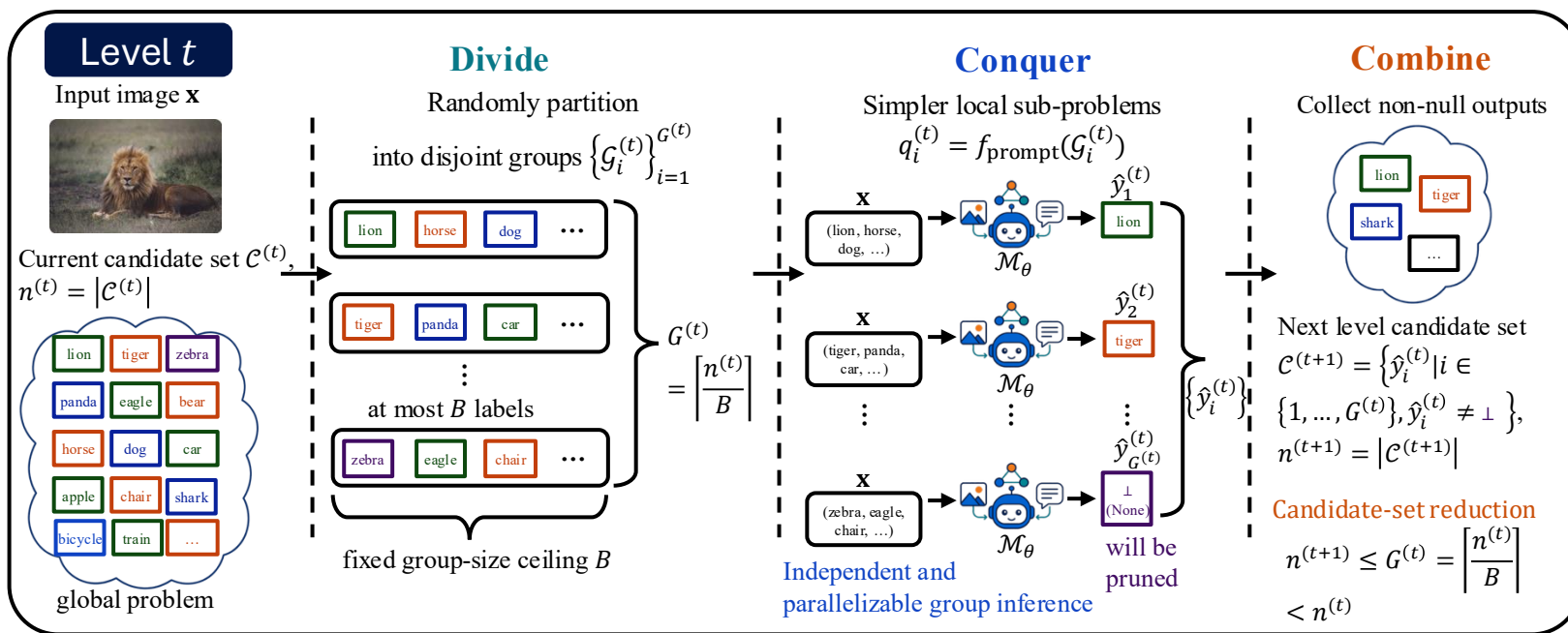


Divide-and-Conquer Inference Method for MLLM-Based Image Classification



Recursion termination check

$$n^{(t+1)} = 0$$

(empty)

$$\hat{y} = \perp$$

Return $\perp$ (no model call)

$$n^{(t+1)} = 1$$

lion

$$\hat{y} = \text{lion}$$

Return sole candidate

$$2 \leq n^{(t+1)} \leq B$$

Final local inference

$\mathbf{x}$ (lion, tiger, apple, ...)

$\mathcal{M}_\theta$

$$\hat{y} = \text{lion}$$

One final model call

$$n^{(t+1)} > B$$

recurse to next level

Level $t+1$

Input image $\mathbf{x}$

Current candidate set $\mathcal{C}^{(t+1)}$, $n^{(t+1)} = |\mathcal{C}^{(t+1)}|$

lion, tiger, shark, ...

Reduced candidate set

Divide

Randomly partition

into disjoint groups $\{\mathcal{G}_i^{(t+1)}\}_{i=1}^{G^{(t+1)}}$

lion, bear, shark, ...

at most B labels

bicycle, apple, tiger, ...

fixed group-size ceiling B

$$G^{(t+1)} = \left\lceil \frac{n^{(t+1)}}{B} \right\rceil$$

Conquer

Simpler local sub-problems

$q_i^{(t+1)} = f_{\text{prompt}}(\mathcal{G}_i^{(t+1)})$

$\mathbf{x}$ (lion, bear, shark, ...)

$\mathcal{M}_\theta$

$$\hat{y}_1^{(t+1)}$$

$$\hat{y}_{G^{(t+1)}}^{(t+1)}$$

out-of-group labels

Combine

Collect non-null outputs

lion, ...

Next level candidate set

$\mathcal{C}^{(t+2)} = \{\hat{y}_i^{(t+1)} | i \in \{1, \dots, G^{(t+1)}\}, \hat{y}_i^{(t+1)} \neq \perp\}$

$n^{(t+2)} = |\mathcal{C}^{(t+2)}|$

Candidate-set reduction

$n^{(t+2)} \leq G^{(t+1)} = \left\lceil \frac{n^{(t+1)}}{B} \right\rceil < n^{(t+1)}$

Recursive difficulty reduction

Monolithic

One difficult global decision

DCI

Solve simpler local sub-problems



Training-free



Boost Accuracy



Reduce inference latency in large-scale image classification



Shared frozen MLLM across all levels